

Topic 1.4 – Energetics

Practical work to measure energy changes helps students to understand the ideas of uncertainty in measurements and evaluate their results in terms of systematic and random errors.

1.4d	Students need to consider why it is necessary to have a standard set of conditions when quoting thermochemical data and the idea that only energy changes can be measured and, consequently, some fixed points are needed (such as the standard heat of formation of elements being defined as 0 Jmol⁻¹). Examples might include the energy generated by burning different fuels. HSW 1, HSW 3
1.4e	Hess's law is an example of a mathematical model that can be used to bring together data and allow it to be used to calculate energy changes that cannot be measured directly such as standard heat of formation of hydrocarbon fuels. HSW 1